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WHEELS INCLUDING A COATING LAYER AND METHODS FOR MAKING THE SAME

Abstract

A wheel includes a hub portion configured to rotate about a rotational axis. A plurality of blades extends radially outward from the hub portion. Each blade of the plurality of blades includes a leading edge and a trailing edge. The hub portion and the plurality of blades include a substrate metal. The substrate metal of the plurality of blades has coated directly thereon a coating layer that has a hardness of about 800 HV or greater and that includes electroless nickel-phosphorous having a phosphorous content of from about 2 to about 4 wt. %.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is related to and claims all available benefit of Indian Provisional Patent Application IPA: 202411014911 filed Feb. 29, 2024, the entire contents of which are herein incorporated by reference.

TECHNICAL FIELD

[0002] The present disclosure generally relates to wheels for turbo-devices, for example, turbochargers, turbines or turbomachines, and/or the like. More particularly, the present disclosure relates to wheels having a coating layer formed thereon and methods for making the same.

BACKGROUND

[0003] Turbo-devices can be used in a variety of applications. For example, turbochargers for gasoline and diesel internal combustion engines are devices known in the art that are used for pressurizing or boosting the intake air stream, routed to a combustion chamber of the engine, by using the heat and volumetric flow of exhaust gas exiting the engine. Another example includes turbines or turbomachines for fuel cells. The turbine may be operatively connected to a fuel cell system and may be configured as an e-charger, electric turbocharger, or other turbo-device for the fuel cell.

[0004] In the case of turbochargers for internal combustion engines, the exhaust gas exiting the engine is routed into a turbine housing of a turbocharger in a manner that causes an exhaust gas-driven turbine wheel to spin within the housing. The exhaust gas-driven turbine wheel is mounted onto one end of a shaft that is common to a radial air compressor mounted onto an opposite end of the shaft and housed in a compressor housing. Thus, rotary action of the turbine wheel also causes the air compressor to spin within a compressor housing of the turbocharger that is separate from the turbine housing. The spinning action of the air compressor causes intake air to enter the compressor housing and be pressurized or boosted to a desired amount before it is mixed with fuel and combusted within the engine combustion chamber.

[0005] In recent years, there has been increasing pressure in the form of governmental legislation to reduce internal combustion engine emissions, such as NO_x and particulate matter (PM). Oxides of nitrogen (NO_x) may be formed when temperatures in the combustion chamber are about 2500° F. or hotter. At these elevated temperatures, the nitrogen and oxygen in the combustion chamber may chemically combine to form nitrous oxides.

[0006] Exhaust gas recirculation (EGR) is a method that has been used to reduce the level of NO_x in exhaust gases. In EGR systems, some of the exhaust gases that would otherwise be discharged into the environment are recirculated into the intake stream. The recirculated exhaust gases have already combusted and have a significantly lower oxygen content, so they do not burn again when they are recirculated. The exhaust gases may displace some of the normal intake charge. As a result, the combustion process may be cooler by several hundred degrees so that NO_x formation may be reduced.

[0007] The use of EGR, however, results in an increased amount of water that is condensed out of the recirculated exhaust gases. The amount of water that is condensed may depend, for example, on temperature, humidity, and operating speed of the engine. When present, the condensed water droplets in the intake stream are passed through an inlet and impact the spinning compressor wheel, and as a result, an erosive effect may be observed over time. This can cause the components to

prematurely fail.

[0008] Similarly in turbines for fuel cells, when present, condensed water droplets in the intake stream are passed through an inlet and impact the spinning fuel cell turbine wheel, and as a result, an erosive effect may also be observed over time. As a result, such components as well may prematurely fail.

[0009] Accordingly, it is desirable to provide wheels for turbo-devices that are able to withstand the erosive effects of water droplets, without requiring the use of heavier and relatively expensive materials. Furthermore, other desirable features and characteristics of the inventive subject matter will become apparent from the subsequent detailed description of the inventive subject matter and the appended claims, taken in conjunction with the accompanying drawings and this background of the inventive subject matter.

BRIEF SUMMARY

[0010] Wheels having a coating layer formed thereon and methods for making the same, are disclosed herein.

[0011] In an exemplary embodiment, a wheel includes a hub portion configured to rotate about a rotational axis. A plurality of blades extends radially outward from the hub portion. Each blade of the plurality of blades includes a leading edge and a trailing edge. The hub portion and the plurality of blades include a substrate metal. The substrate metal of the plurality of blades has coated directly thereon a coating layer that has a hardness of about 800 HV or greater and that includes electroless nickel-phosphorous having a phosphorous content of from about 2 to about 4 wt. %.

[0012] In another exemplary embodiment, a wheel includes a hub portion configured to rotate about a rotational axis. A plurality of blades extends radially outward from the hub portion. Each blade of the plurality of blades includes a leading edge and a trailing edge. The hub portion and the plurality of blades include a substrate metal. The substrate metal includes aluminum or an alloy thereof. The substrate metal of the plurality of blades has coated directly thereon a coating layer that has a hardness of from about 800 to about 950 HV and that includes electroless nickel-phosphorous having a phosphorous content of from about 2 to about 4 wt. %.

[0013] In another exemplary embodiment, a method for making a wheel includes providing a substrate wheel. The substrate wheel includes a hub portion configured to rotate about a rotational axis. A plurality of blades extends radially outward from the hub portion. Each blade of the plurality of blades includes a leading edge and a trailing edge. The hub portion and the plurality of blades include a substrate metal. The method further includes forming on the substrate metal a coating layer that includes electroless nickel-phosphorous having a phosphorous content of from about 2 to about 4 wt. %. Forming the coating layer includes immersing the substrate wheel in an electroless nickel-phosphorous plating bath that includes nickel cations and phosphorous oxide anions. The method further includes exposing the substrate wheel including the coating layer to a heat treatment process at heat treating conditions effective to increase a hardness of the coating layer to about 800 HV or greater.

[0014] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] The inventive subject matter will hereinafter be described in conjunction with the following drawing figures, wherein like numerals denote like elements, and wherein:

[0016] FIG. 1 is a perspective view of a wheel operatively disposed in a turbo-device, which is

schematically illustrated, in accordance with some embodiments of the present disclosure; [0017] FIG. 2 is a perspective view of a wheel operatively disposed in a turbo-device, which is schematically illustrated, in accordance with some embodiments of the present disclosure; and [0018] FIG. 3 is a flowchart illustrating a method for making a wheel in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0019] As required, detailed embodiments of the present disclosure are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the disclosure that may be embodied in various and alternative forms. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

[0020] Unless specifically stated or obvious from context, as used herein, the term “about” is understood as within a range of normal tolerance in the art, for example within 2 standard deviations of the mean. “About” can be understood as within 10%, 5%, 1%, 0.5%, 0.1%, 0.05%, or 0.01% of the stated value. “About” can alternatively be understood as implying the exact value stated. Unless otherwise clear from the context, all numerical values provided herein are modified by the term “about.”

[0021] The present disclosure is generally directed to wheels for turbo-devices in which the wheels have a coating layer disposed thereon and methods for making the same. In particular, the present disclosure addresses the aforementioned erosion problem with the use of a relatively low range phosphorous content (e.g., about 2 to about 4 wt. %), electroless nickel-phosphorus as a protective coating layer overlying the wheel's substrate metal formed, for example, of a relatively soft aluminum or aluminum alloy substrate. The coating layer has been heat treated so as to have a relatively high hardness (about 800 HV or greater). This coating layer disposed directly on the soft aluminum substrate provides additional hardness (e.g., about 800 HV or greater) and helps the wheel including particularly the blades to withstand the erosive effects of water droplets, without requiring the use of heavier and relatively expensive materials.

[0022] Referring to FIG. 1, a perspective view of a wheel **10** operatively disposed in a turbo-device **12** is provided. In the illustrated embodiment, the turbo-device **12** is configured as a turbocharger **14** for an internal combustion engine and the wheel **10** is configured as a turbocharger compressor wheel **16**. A non-limiting example of turbochargers for internal combustion engines including turbocharger compressor wheels is described in U.S. Pat. No. 11,566,631, filed on Mar. 29, 2021, which is owned by the assignee of the present application and is hereby incorporated by reference in its entirety for all purposes.

[0023] As illustrated, the wheel **10** is operatively disposed in the turbo-device **12** between an inlet **18** and an outlet **20** to rotate (indicated by single headed arrow **13**) about a rotational axis **22**. The wheel **10** is a radial wheel that includes a hub portion **24** and a plurality of blades **26** that extend radially outward from the hub portion **24**. The blades **26** have a backward curvature rather than being configured to extend in a purely radial blade configuration. Each blade **26** includes a leading edge **28** that is in fluid communication with the inlet **18** and a trailing edge **30** that is in fluid communication with the outlet **20**. The leading edges **28** define the beginning of an intake area for the combined set of blades **26**, extending through the circular paths of roughly the upstream $\frac{1}{3}$ of the blades **26**. The trailing edges **30** define the end of an annular output area for the combined set of blades **26**, extending through the circular paths of roughly the downstream $\frac{1}{3}$ of the blades **26**.

[0024] During operation of the turbo-device **12**, the wheel **10** rotates about the rotational axis **22** and the leading edges **28** receive intake air that passes through the inlet **18** and advances rearwardly (indicated by single headed arrow **32**) along the blades **26** towards the trailing edges **30**. As such, the leading edges **28** are positioned longitudinally forward of the trailing edges **30** of the blades **26**

with respect to the rotational axis **22** and the flow of air **32** along the wheel **10**. As noted above, the wheel **10** is a turbocharger compressor wheel **16** in which the blades **26** are configured to compress the intake air to form compressed or pressurized air. The pressurized air passes from the trailing edges **30** and is ejected out through the outlet **20**.

[0025] In some embodiments, the hub portion **24** and the blades **26** are formed of a substrate metal **36**, such as, aluminum or an aluminum alloy, for example, via a casting and/or machining process. The wheel **10** is provided with a coating layer **34** on and overlying the substrate metal **36** and includes or is formed of electroless nickel-phosphorous. The phosphorous content of the coating layer **34** may be greater than or equal to about 2 wt. %, for example from about 2 to about 4 wt. %, such as from about 2 to about 3 wt. %, for example from about 2 to about 2.6 wt. %. In an exemplary embodiment, the nickel content of the first coating layer **34** is less than or equal to about 98 wt. %, such, from about 96 to about 98 wt. %, such as from about 97 to about 98 wt. %, for example, from about 97.4 to about 98 wt. %.

[0026] In an exemplary embodiment, the coating layer **34** has a thickness from about 20 to about 30 microns, for example from about 22 to about 28 microns. The coating layer **34** may be provided on all or most of the surfaces of the wheel **10**, both forward and rear facing. If the coating layer **34** is not provided on all of the surfaces, the surfaces not coated with the coating layer **34** may include functional surfaces, such as portions of the back facing hub portion **24** about the centerline (axis of rotation **22**) or portions of the forward-facing hub portion **24**.

[0027] In an embodiment, the coating layer **34** has a hardness of about 800 HV or greater. Some embodiments, the coating layer **34** has a hardness of about 800 HV to about 950 HV. As will be discussed in further detail below, in some embodiments, the wheel **10** has been subjected to a heat treating process after deposition of the electroless nickel-phosphorous that forms the coating layer **34** to improve the hardness of the coating layer **34** to enhance erosion resistance.

[0028] Referring to FIG. 2, a wheel **100** that is operatively disposed in a turbo-device **112** between an inlet **118** and an outlet **120** is provided. In the illustrated embodiment, the turbo-device is a turbine **114** for fuel cells and the wheel **100** is configured as a fuel cell turbine wheel **116**. A non-limiting example of turbines for fuel cells including fuel cell turbine wheels is described in U.S. Patent Application Publication No. 2022/0006369, filed on Jul. 1, 2020, which is owned by the assignee of the present application and is hereby incorporated by reference in its entirety for all purposes.

[0029] In particular, the wheel **100** including the rotational axis **122**, the hub portion **124**, the blades **126**, the leading edges **128**, the trailing edges **130**, the substrate metal **136**, and the coating layer **134** are similarly configured to the wheel **10** as discussed above in relation to FIG. 1 including the rotational axis **22**, the hub portion **24**, the blades **26**, the leading edges **28**, the trailing edges **30**, the substrate metal **136**, and the coating layer **134**, respectively, but with the exception that the blades **126** are configured to expand the intake air received from the inlet **118**, along the airflow direction **132** towards the trailing edges **130** to form an expanded or depressurized air. The expanded or depressurized air passes from the trailing edges **130** and is ejected out through the outlet **120**.

[0030] Referring to FIG. 3, the compressor wheel **10**, **100** as discussed above may be made in accordance with a method **200** as illustrated in the flowchart. The method **200** includes a step **202** of providing a substrate wheel formed of a substrate metal, specifically a wheel made of aluminum (or alloy thereof) in the configuration discussed above in relation to FIGS. 1-2, with the exception of the coating layer. The substrate wheel may be manufactured using conventional manufacturing processes, such as casting and/or machining, or the like.

[0031] The method **200** continues with a step **204** of forming (e.g., via depositing) an electroless nickel-phosphorous coating layer onto the substrate metal of the substrate wheel. Electroless nickel-phosphorus plating is a chemical process that deposits an even layer of nickel-phosphorus alloy on the surface of the substrate metal. The process involves dipping the substrate wheel in a

water solution containing a nickel salt and a phosphorus-containing reducing agent, for example a hypophosphite salt. The concentration of the phosphorous-containing reducing agent is selected so as to achieve a phosphorous amount in the coating layer greater than or equal to about 2 wt. %, for example about 2 to about 4 wt. %, as described above, and a nickel amount in the coating layer less than or equal to about 98 wt. %, for example, about 96 to about 98 wt. %, as described above. The reduction of the metal cations in solution to metallic form is achieved by purely chemical means, through an autocatalytic reaction. Before plating, the surface of the substrate may be cleaned. Cleaning may be achieved by a series of chemical baths, including non-polar solvents to remove oils and greases, as well as acids and alkalis to remove oxides, insoluble organics, and other surface contaminants. Further, functional portions of the substrate metal, as described above, may be optionally masked. Ingredients of the electroless nickel plating bath include a source of nickel cations Ni^{2+} , for example nickel sulfate and a suitable reducing agent, such as hypophosphite H_2PO_3^- . The plating bath may further include complexing agents, such as carboxylic acids or amines; stabilizers, such as lead salts or sulfur compounds; buffers; surfactants; and accelerators. In an exemplary embodiment, the plating bath has a pH of about 5 to about 8, for example about 5 to about 7. The plating process is controlled with temperature and time to achieve a desired uniform thickness of about 20 to about 30 microns, as described above. In an exemplary embodiment, the substrate wheel is immersed in the plating bath for a time of about 130 to about 170 minutes. Once Ni—P plating is complete, the substrate metal, now having the coating layer plated thereon, may be rinsed to remove any residues from the plating process, and the masking (if any) may be removed.

[0032] The method **200** continues with a step **206** by exposing the substrate wheel including the coating layer to a heat treating process at heat treating conditions effective to improve the hardness of the coating layer **34** to enhance erosion resistance. As described above, the heat treating process increases the hardness of the coating layer to about 800 HV or greater, for example, about 800 HV to about 950 HV. In an exemplary embodiment, step **206** includes exposing the substrate wheel including the coating layer to a temperature of about 200° C. to about 230° C. for a time of about 1 to about 4 hours. The method **200** may further include performing various finishing processes, such as final cleaning, polishing, and machining in addition to the heat treating process. The result is a wheel **10**, **100** in accordance with that described above in connection with FIGS. **1-2**.

[0033] While at least one exemplary embodiment has been presented in the foregoing detailed description of the inventive subject matter, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the inventive subject matter in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing an exemplary embodiment of the inventive subject matter. It being understood that various changes may be made in the function and arrangement of elements described in an exemplary embodiment without departing from the scope of the inventive subject matter as set forth in the appended claims.

Claims

1. A wheel comprising: a hub portion configured to rotate about a rotational axis; and a plurality of blades extending radially outward from the hub portion, wherein each blade of the plurality of blades comprises a leading edge and a trailing edge, wherein the hub portion and the plurality of blades comprise a substrate metal, wherein the substrate metal of the plurality of blades has coated directly thereon, without an intermediate chemical conversion or phosphate layer, a coating layer configured to enhance erosion resistance against condensed water droplets in exhaust gas recirculation or fuel cell environments, the coating comprising: a hardness of about 800 HV or greater, electroless nickel-phosphorous having a phosphorous content of from about 2 to about 4

wt. %, and a thickness of from about 20 to about 30 microns.

2. The wheel of claim 1, wherein the coating layer comprises the electroless nickel-phosphorous having the phosphorous content of from about 2 to about 3 wt. %.

3. The wheel of claim 1, wherein the coating layer comprises the electroless nickel-phosphorous having the phosphorous content of from about 2 to about 2.6 wt. %.

4. The wheel of claim 1, wherein the coating layer comprises the electroless nickel-phosphorous having a nickel content of about 96 to about 98 wt. %.

5. The wheel of claim 1, wherein the hardness of the coating layer is from about 800 to about 950 HV.

6. (canceled)

7. The wheel of claim 1, wherein the coating layer extends overlaying the substrate metal of the hub portion.

8. The wheel of claim 1, wherein the substrate metal comprises aluminum or an alloy thereof.

9. The wheel of claim 1, wherein the wheel is configured as a turbocharger compressor wheel or a fuel cell turbine wheel.

10. A wheel comprising: a hub portion configured to rotate about a rotational axis; and a plurality of blades extending radially outward from the hub portion, wherein each blade of the plurality of blades comprises a leading edge and a trailing edge, wherein the hub portion and the plurality of blades comprise a substrate metal that comprises aluminum or an alloy thereof, wherein the substrate metal of the plurality of blades has coated directly thereon, without an intermediate chemical conversion or phosphate layer, a coating layer configured to enhance erosion resistance, the erosion caused by condensed water droplets, the coating comprising a hardness of from about 800 to about 950 HV, electroless nickel-phosphorous having a phosphorous content of from about 2 to about 4 wt. %, and a thickness of from about 20 to about 30 microns.

11. A method for making a wheel, the method comprising: providing a substrate wheel that comprises: a hub portion configured to rotate about a rotational axis, and a plurality of blades extending radially outward from the hub portion, wherein each blade of the plurality of blades comprises a leading edge and a trailing edge, wherein the hub portion and the plurality of blades comprise a substrate metal; forming on the substrate metal a coating layer comprising electroless nickel-phosphorous having a phosphorous content of from about 2 to about 4 wt. %, wherein forming the coating layer comprises immersing the substrate wheel in an electroless nickel-phosphorous plating bath comprising nickel cations and phosphorous oxide anions; and exposing the substrate wheel including the coating layer to a heat treatment process at heat treating conditions effective to increase a hardness of the coating layer to about 800 HV or greater.

12. The method of claim 11, wherein exposing includes the heat treating conditions including a temperature of about 200° C. to about 230° C. for a time of about 1 to about 4 hours.

13. The method of claim 11, wherein providing the substrate wheel comprises providing the substrate wheel comprising the substrate metal that comprises aluminum or an alloy thereof.

14. The method of claim 13, wherein forming the coating layer comprises immersing the substrate wheel that comprises aluminum or the alloy thereof in the electroless nickel-phosphorous plating bath comprising nickel cations and phosphorous oxide anions at a pH of from about 5 to about 8.

15. The method of claim 14, wherein the pH of the electroless nickel-phosphorous plating bath is from about 5 to about 7.

16. The method of claim 11, wherein forming the coating layer comprises immersing the substrate wheel in the electroless nickel-phosphorous plating bath for a time of about 130 to about 170 minutes.

17. The method of claim 11, wherein forming the coating layer comprises forming the coating layer having a thickness of from about 20 to about 30 microns.

18. The method of claim 11, wherein exposing the substrate wheel comprises exposing the coating layer to the heat treatment process at the heat treating conditions to increase the hardness of the

coating layer to about 800 to about 950 HV.

19. The method of claim 11, wherein forming the coating layer comprises forming the coating layer that comprises the electroless nickel-phosphorous having the phosphorous content of from about 2 to about 3 wt. %.

20. The method of claim 11, wherein forming the coating layer comprises forming the coating layer that comprises the electroless nickel-phosphorous having the phosphorous content of from about 2 to about 2.6 wt. %.
